CCC-C-444D
March 14, 1973
SUPERSEDING
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June 4, 1969

FEDERAL SPECIFICATION

CLOTH, JEAN, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton jean cloth.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.2).

Class 1 - Bleached.
Class 2 - Dyed.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276	-	Thread, Cotton.
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4	-	Glossary of Fabric Imperfections.
Fed. Std. No. 123	-	Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 191	-	Textile Test Methods.

FSC 8305

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the bleached or dyed cloth shall match the sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, drawn and spun into single yarns.

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3.3 Color.

3.3.1 Dyed cloth. Unless otherwise specified the color shall be bleached or dyed Grayish Green 180, as specified (see 6.2). The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method, when tested as specified in 4.4.

3.3.2 White cloth. When white is specified, the cloth shall be bleached white to match the standard sample (see 6.3). The cloth shall be fully bleached and may be supplemented with fluorescent optical brighteners with a resulting peak emission on the blue violet. The white finished cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.4. When no standard sample is available the discoloration shall be no worse than the rating of "fair" when tested as specified in 4.4.

3.3.3 Matching. When a standard sample is available, the dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.4 Colorfastness. The dyed and finished cloth shall show colorfastness to laundering (after 3 cycles), perspiration, crocking and light equal to or better than the standard sample (see 6.3). When a standard sample is not available or not designated as applicable to colorfastness, the dyed cloth shall show "good" colorfastness to laundering (after 3 cycles), perspiration, light, and no more crocking than Munsell value 8.5 when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Class	Weight per square yard (minimum) Ounces	Yarns per inch (minimum) Warp	Yarns per inch (minimum) Filling	Breaking strength (minimum) (pounds) Warp	Breaking strength (minimum) (pounds) Filling	Weave
1	4.7	98	60	80	48	2/1 twill
2	4.5	98	60	80	48	

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3.4.1 Width. The width shall be as specified and shall be the minimum width inclusive of selvages (see 6.2).

3.5 Finish. The cloth shall be singed, desized, and bleached or dyed, as required.

3.6 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 pH. The pH value of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.8 Shrinkage. The cloth shall be preshrunk and the residual shrinkage shall not exceed 1.0 percent in either the direction of the warp or of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of no less than 85 percent when tested as specified in 4.4.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in one continuous length of not less than 50 yards and shall be put-up in bolts or flat folded pieces as specified in PPP-P-1134.

3.11 Fiber identification. Each piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.12 Face marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.13 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown in figure 1 of Fed. Std. No. 4, shall be scored. No linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed the following point values:

35.0 points for white cloth.
30.0 points for dyed cloth.

The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceed the following values:

53.0 points for white cloth.
45.0 points for dyed cloth.

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If one piece exceeds the point level, a second sample of 20 pieces shall be selected for individual piece quality examination only. The lot shall be unacceptable if one or more pieces in the second sample exceed the point level. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards.}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point.
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points.
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points.
For defects exceeding 9 inches in any dimension	- four points.

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

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4.2.2.3 Examination for shade. During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample, off-shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for identification of preshrinkage process, compliance with the Textile Fiber Products Identification Act, and face markings. During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Preshrinkage process identified by name or trademark on cloth or ticket.
Not labeled ticketed or invoiced in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.
Face stamping missing from either or both ends.
Face stamping on wrong side.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191 wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 4-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.2.1	1200 1/
Presence of labile sulfur	3.3.1	2020
Discoloration (bleached cloth)	3.3.2	5660 2/ 3/
Colorfastness to:		
Laundering (after 3 cycles)	3.3.4	5610 4/ 5/
Perspiration	3.3.4	5680
Light	3.3.4	5660
Crocking	3.3.4	5651
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 2/
Singed	3.5	Visual 1/
Bleached (Class 1)	3.3.2 and 3.5	Visual 1/
Nonfibrous material	3.6	2611
pH	3.7	2811
Shrinkage	3.8	5550
Seam efficiency	3.9	5110 6/

1/ Unless otherwise specified, a certificate of compliance shall be submitted by the supplier and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

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3/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration shall be considered satisfactory if it is not worse than the rating of "fair" indicated in Method 5660.

4/ The specimens must be dried after each of the 3 laundering cycles.

5/ On the color transfer cloth evaluation, only the stain on the cotton fiber of the color transfer cloth shall be evaluated.

6/ The needle shall measure 0.040 inch (± 0.001 inch) across the blade at the eye. The cotton thread shall be type 1A1, ticket No. 40, 3 ply in the needle and ticket No. 60, 3 ply in the loopers, conforming to V-T-276.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking.

5.3.1 Civil requirements. In addition to any special marking by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of clothing and equipage items. The dyed cloth is utilized for medical clothing items.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 3.3).
(c) Width of cloth required (see 3.4).
(d) When length of piece is other than specified (see 3.10).
(e) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. Procuring officers, or suppliers, may obtain a standard sample from the procuring activity issuing the invitation for bids.

6.4 Dye formulation. Satisfactory results for achieving the proper shade for Grayish Green 180 and colorfastness have been obtained with an appropriate combination of the following dyes:

Vat Black 18
Vat Green 3, C.I. 69500
Vat Blue 53, C.I. 71205

6.4.1 Standard or guide samples of the Grayish Green 180 color may be obtained from the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101.

6.5 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A requirements shall apply.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Review activity:

Army - MD

User activity:

Navy - MC

Preparing activity:

Army - GL

Civil Agency Interest:

GSA-FSS
HEW-HSM
INT-BPA
VA-DSM
JUS-FPI

Project No. 8305-0095

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION CCC-C-444D CLOTH, JEAN, COTTON

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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